A HALF-CONSTANT UPPER BOUND FOR CHOWLA'S COSINE PROBLEM

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ABSTRACT. Let $\lambda(n)$ be the infimum of $-\min_x \sum_{a\in A}\cos(ax)$ over all sets A of n positive integers. The classical Sidon-difference construction gives $\lambda(n) \le (1+o(1))\sqrt{n/2}$. We construct sets for which the negative excursion of the cosine sum has leading constant $1/2$: $\lambda(n) \le \frac{1}{2}\sqrt{n} + O(n^{1/4}\sqrt{\log n})$, so that $\limsup_{n\to\infty} \lambda(n)/\sqrt{n} \le 1/2$. The construction combines a two-sided Sidon square identity with a Singer difference set, a Fejér-kernel majorant, a parity-biased deletion, and an exact-cardinality probabilistic selection. We also record an L^4 criterion and a multiplicity bound showing what any construction with a smaller leading constant must achieve. In the opposite direction we carry the L^1 norm through Bedert's boundary estimate, obtaining it with an explicit constant and with one factor of $-\min f_A$ replaced by $\|f_A\|_1$, where $f_A(x) = \sum_{a\in A}\cos(ax)$, and deduce two lower bounds: $-\min f_A \ge cn^{1/4}$ for every n-element set $A \subset [1, Cn]$, and $(-\min f_A)^3\|f_A\|_1^2 \ge cn$ for every n-element $A \subset \mathbb{N}$. The second recovers the exponent $1/5$ of Bedert and Shankar and improves it whenever $\|f_A\|_1$ is small compared with $-\min f_A$. Numerical values of the construction are reported up to $|A| = 25440$.

1. INTRODUCTION

For a finite set $A \subset \mathbb{N}$ write

$$f_A(x) = \sum_{a\in A}\cos(ax),$$

and define

$$\lambda(n) = \inf_{\substack{A\subset\mathbb{N}\\ |A|=n}} \Big(-\min_{x\in\mathbb{R}} f_A(x)\Big),$$

the least possible depth of the negative excursion of a cosine sum with n frequencies. Since f_A has mean zero and $f_A(0) = n$, one has $\lambda(n) > 0$. Ankeny and Chowla [3] asked whether $\lambda(n) \to \infty$; Uchiyama and Uchiyama [16] proved this using Cohen's work on the Littlewood L^1 problem, and

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the resolution of that problem by Konyagin [7] and McGehee, Pigno and Smith [8] gives $\lambda(n) \gg \log n$. Bourgain [2] and Ruzsa [12] obtained $\lambda(n) \geq \exp(c\sqrt{\log n})$, and polynomial lower bounds were proved recently: $\lambda(n) \geq n^{1/10-o(1)}$ by Jin, Milojević, Tomon and Zhang [6] and $\lambda(n) \geq n^{1/5-o(1)}$ by Bedert [1], sharpened to $\lambda(n) \gg n^{1/5}$ by Shankar [14]; the spectral method of [6] has since been refined by Yip [17] to $\lambda(n) \gg n^{1/6.43}$. Chowla [4] conjectured that $\lambda(n) \asymp \sqrt{n}$. Explicit values are known only for $n = 2, 3$ [9].

In the other direction, the standard construction, and the only one we are aware of, is the Sidon-difference construction. A set $S \subset \mathbb{Z}$ is *Sidon* if every nonzero difference has at most one ordered representation, equivalently if $s_1 - s_2 = s_3 - s_4$ and $s_1 \neq s_2$ force $(s_1, s_2) = (s_3, s_4)$. If B is a Sidon set of size k and $A = \{b - b' > 0 : b, b' \in B\}$, then $|A| = \binom{k}{2}$ and

$$f_A(x) = \tfrac{1}{2}\Big(\Big|\sum_{b\in B} e^{ibx}\Big|^2 - k\Big) \geq -\frac{k}{2},$$

so that $\lambda(\binom{k}{2}) \leq k/2$, that is,

$$\lambda(m) \leq (1 + o(1))\sqrt{m/2} \qquad (m = \tbinom{k}{2} \to \infty). \tag{1.1}$$

This gives the leading constant $1/\sqrt{2} \approx 0.707$ along the cardinalities $\binom{k}{2}$. The explicit form recorded by Mercer [10, Fact 2.1], valid for every m, is $\lambda(m) \leq \sqrt{2m} + 1/2$; the larger constant there comes from padding the difference set with up to $\sqrt{2m}$ arbitrary frequencies, each bounded trivially. Recent papers quote the construction only as $\lambda(n) \ll \sqrt{n}$ with an unspecified or generous constant (for instance $10\sqrt{n}$ in [1]). We are not aware of any published construction with leading constant below $1/\sqrt{2}$.

In this paper we sharpen the leading constant to $1/2$. The idea is to use sums as well as differences of a Sidon set: for a real cosine sum ρ with k frequencies, ρ^2 produces $k(k-1)$ distinct nondiagonal frequencies while the diagonal contribution is only $k/2$. The price is a correction term (the sum g below) that is not controlled by positivity alone, and the bulk of the paper is the construction of a Sidon set for which this correction is negligible at every point of the circle.

Results. For $S = \{s_1, \dots, s_k\} \subset \mathbb{Z}$ define

$$A(S) = \{3|s_i - s_j| : i < j\} \cup \{|3(s_i + s_j) + 1| : i < j\},$$

$$\rho(x) = \sum_{j=1}^{k} \cos\big((3s_j + \tfrac{1}{2})x\big), \qquad g(x) = \sum_{j=1}^{k} \cos\big((6s_j + 1)x\big).$$

Proposition 1.1 (Two-sided Sidon identity)**.** *If $S \subset \mathbb{Z}$ is Sidon and $|S| = k \geq 2$, then $A(S) \subset \mathbb{N}$ has $|A(S)| = k(k-1)$, and for every real x,*

$$f_{A(S)}(x) = \rho(x)^2 - \frac{k}{2} - \frac{1}{2}g(x).$$

Theorem 1.2 (Base cardinalities)**.** *For every sufficiently large integer k there exists a set $A \subset \mathbb{N}$ with $|A| = k(k-1)$ such that*

$$\min_x f_A(x) \geq -\frac{k}{2} - O\big(\sqrt{k \log k}\big).$$

Equivalently, $\lambda(k(k-1)) \leq k/2 + O(\sqrt{k \log k})$.

Theorem 1.3 (Half-constant upper bound)**.** *As $n \to \infty$,*

$$\lambda(n) \leq \frac{1}{2}\sqrt{n} + O\big(n^{1/4}\sqrt{\log n}\big), \qquad \textit{so that} \qquad \limsup_{n\to\infty} \frac{\lambda(n)}{\sqrt{n}} \leq \frac{1}{2}.$$

Section 8 explains why $1/2$ is the natural endpoint of this method: an L^4 identity (Proposition 8.1) and a multiplicity bound (Proposition 8.2) show that in any construction of the form "square of a trigonometric polynomial with coefficients of constant modulus, minus an analytically absorbed correction", a leading constant below $1/2$ requires every frequency to have at least three representations as a difference of the support, whereas the only symmetry of $\mathbb{Z}$ produces two. Section 9 concerns the opposite direction. The boundary estimate of Bedert [1, Proposition 7.3] bounds, for a symmetric set $A' \subset \mathbb{Z} \setminus \{0\}$ whose exponential sum is at least $-K$, the number of maximal arithmetic progressions in A' with a given common difference and at least two terms by $O(K^4)$. We reproduce his argument with the L^1 norm of the exponential sum carried through and obtain the bound $O(K^3 \|\widehat{\mathbf{1}_{A'}}\|_1)$ with an explicit constant (Proposition 9.1). Combined with an elementary count of three-term progressions this gives $-\min f_A \geq cn^{1/4}$ for every $A \subset [1, Cn]$ with $|A| = n$ (Proposition 9.7), a class that contains the sets of Theorem 1.2, and for which the finite-group results of Sanders [13] give a polynomial bound with an exponent that decays with C. Combined with Shankar's total-boundary identity [14] and a third-moment inequality for the positive part of f_A it gives $(-\min f_A)^3 \|f_A\|_1^2 \geq cn$ for every n-element $A \subset \mathbb{N}$ (Theorem 9.11), which contains the bound $-\min f_A \gg n^{1/5}$ of [1, 14] because $\|f_A\|_1 \leq -2\min f_A$, and improves it when the L^1 norm is smaller; a companion form of Ruzsa's lemma bounds arithmetic progressions in A by $64(-\min f_A)\|f_A\|_1 + 6$ (Corollary 9.13). Section 10 reports numerical values of the construction.

Organisation. Section 2 proves Proposition 1.1. Section 3 sets up the Singer difference set and the Fejér-kernel majorant $m(y)$. Section 4 isolates a parity block of elements near the centre of the difference set. Section 5 carries out the exact-cardinality probabilistic thinning. Section 6 performs the three-zone estimate and proves Theorem 1.2. Section 7 passes from the base cardinalities $k(k-1)$ to every cardinality n and proves Theorem 1.3. Section 8 contains the L^4 criterion, Section 9 the lower bounds, and Section 10 the numerical results.

2. The square identity

Proposition 2.1 (Two-sided Sidon identity)**.** *If $S \subset \mathbb{Z}$ is Sidon and $|S| = k \geq 2$, then $A(S) \subset \mathbb{N}$ has*

$$|A(S)| = k(k-1),$$

and for every real x,

$$f_{A(S)}(x) = \rho(x)^2 - \frac{k}{2} - \frac{1}{2}g(x).$$

Proof. The frequencies $3|s_i - s_j|$ are divisible by 3, whereas $|3(s_i+s_j)+1|$ is congruent to ± 1 modulo 3, so the two families are disjoint, and all frequencies are positive. The Sidon property gives distinct difference frequencies. If

$$|3(s_i + s_j) + 1| = |3(s_a + s_b) + 1|,$$

the negative-sign alternative would imply $3(s_i+s_j+s_a+s_b) = -2$, which is impossible. Thus $s_i + s_j = s_a + s_b$, and the Sidon property implies $\{i, j\} = \{a, b\}$. Hence each family has $\binom{k}{2}$ elements.

Writing $\alpha_j = 3s_j + \frac{1}{2}$ and expanding,

$$\begin{aligned}\rho(x)^2 &= \sum_j \cos^2(\alpha_j x) + 2\sum_{i<j} \cos(\alpha_i x)\cos(\alpha_j x) \\ &= \frac{k}{2} + \frac{1}{2}\sum_j \cos\big((6s_j+1)x\big) + \sum_{i<j} \cos\big(3(s_i - s_j)x\big) \\ &\quad + \sum_{i<j} \cos\big((3(s_i+s_j)+1)x\big).\end{aligned}$$

Since cosine is even and all frequencies above occur exactly once, the last two sums are $f_{A(S)}(x)$. Rearranging proves the identity. □

The significance of Proposition 2.1 is that $k(k-1) \sim k^2$ positive frequencies are created while the unavoidable diagonal contribution is only $k/2$. The remainder of the proof constructs S so that $\rho^2 - g/2 \geq -o(k)$ uniformly in x. Note that positivity of ρ^2 alone does not suffice: at a zero

x_0 of ρ one has $f_{A(S)}(x_0) = -k/2 - g(x_0)/2$, and g is a cosine sum with k terms.

3. Singer sets and Fejér smoothing

Let q be a prime and

$$v = q^2 + q + 1.$$

We use a Singer difference set [15] $D \subset \mathbb{Z}/v\mathbb{Z}$ of size $q+1$ with the perfect-difference property: every nonzero residue modulo v has a unique representation $d - d'$ with $d, d' \in D$, $d \neq d'$. Choose representatives $D \subset \{0, \dots, v-1\}$ and put

$$c = \frac{v-1}{2}, \qquad L = c + 1 = \frac{v+1}{2}.$$

Every subset of $D - c \subset \mathbb{Z}$ is Sidon over the integers, since a coincidence of differences over $\mathbb{Z}$ would be a coincidence modulo v.

Define the tent weight

$$w(n) = 1 - \frac{|n - c|}{L}, \qquad 0 \le n \le v - 1,$$

and the Fejér kernel

$$F_M(y) = \frac{1}{M}\Big(\frac{\sin(My/2)}{\sin(y/2)}\Big)^2.$$

Set

$$m(y) = \frac{q+1}{v} F_L(y), \qquad \kappa = m(0) = \frac{(q+1)L}{v}.$$

Thus

$$\frac{q+1}{2} \le \kappa \le \frac{q+2}{2}.$$

We shall use

$$0 \le m(y) \le \kappa, \qquad m(y) \le \frac{q+1}{vL} \cdot \frac{1}{\sin^2(y/2)}, \tag{3.1}$$

and, for $|u| < \pi$,

$$m(2u) \le \frac{m(u)}{\cos^2(u/2)}. \tag{3.2}$$

The last inequality follows from $\sin^2(Lu) = 4\sin^2(Lu/2)\cos^2(Lu/2) \le 4\sin^2(Lu/2)$ and $\sin^2 u = 4\sin^2(u/2)\cos^2(u/2)$.

Lemma 3.1 (Singer flat spectrum)**.** *For $j \not\equiv 0 \pmod v$,*

$$\Big|\sum_{d \in D} e^{-2\pi i j d/v}\Big|^2 = q.$$

Consequently, for every $\phi : \{0, \dots, v-1\} \to \mathbb{C}$, if

$$\Phi(j) = \sum_{n=0}^{v-1} \phi(n) e^{2\pi i j n / v},$$

then

$$\Big| \sum_{d \in D} \phi(d) - \frac{q+1}{v} \Phi(0) \Big| \le \frac{\sqrt{q}}{v} \sum_{j=1}^{v-1} |\Phi(j)|.$$

Proof. The perfect-difference property gives

$$\Big| \sum_{d \in D} e^{-2\pi i j d / v} \Big|^2 = q + 1 + \sum_{u=1}^{v-1} e^{2\pi i j u / v} = q.$$

By Fourier inversion $\phi(n) = v^{-1} \sum_j \Phi(j) e^{-2\pi i j n / v}$, so

$$\sum_{d \in D} \phi(d) = \frac{1}{v} \sum_{j=0}^{v-1} \Phi(j) \sum_{d \in D} e^{-2\pi i j d / v},$$

and the second assertion follows from the first and the triangle inequality. □

Lemma 3.2 (Tent-weighted Singer sum)**.** *For every real y,*

$$\sum_{d \in D} w(d) e^{i(d-c)y} = m(y) + R(y), \qquad |R(y)| \le \sqrt{q}.$$

Proof. The Fourier series of the tent is

$$\sum_{n=0}^{v-1} w(n) e^{iny} = e^{icy} F_L(y),$$

since $n - c$ ranges over $|n - c| \le L - 1$ and $\sum_{|m|<L} (1 - |m|/L) e^{imy} = F_L(y)$. Apply Lemma 3.1 with $\phi(n) = w(n) e^{i(n-c)y}$. Then $\Phi(j) = e^{2\pi i j c / v} F_L(y + 2\pi j / v)$, so $\frac{q+1}{v} \Phi(0) = m(y)$ and $|\Phi(j)| = F_L(y + 2\pi j / v) \ge 0$. The discrete aliasing identity

$$\sum_{j=0}^{v-1} F_L(y + 2\pi j / v) = v,$$

valid because no nonzero $|m| < L$ is divisible by v, shows that $\sum_{j \ne 0} |\Phi(j)| \le v$, and the total nonzero-frequency error is at most $\sqrt{q}$. □

4. A parity block near the centre

For $0 \le r \le c$ let

$$N_{\mathrm{odd}}(r) = \#\{d \in D : |d - c| \le r,\ d - c \text{ odd}\}.$$

Lemma 4.1 (Odd-offset counting)**.** *Let $1 \le h \le a$ and $a + h \le c$. Then*

$$N_{\mathrm{odd}}(a+h) \ge \frac{q+1}{v}\Big(a - \frac{1}{4h}\Big) - \frac{a}{h}\sqrt{q}.$$

Proof. Define the trapezoidal cutoff

$$\psi(u) = \frac{\max(0, a+h-|u|) - \max(0, a-h-|u|)}{2h}.$$

Then $0 \le \psi \le 1$, it is supported on $|u| < a+h$, it equals 1 for $|u| \le a-h$, and

$$\widehat{\psi}(\theta) := \sum_{u\in\mathbb{Z}} \psi(u) e^{iu\theta} = \frac{(a+h)F_{a+h}(\theta) - (a-h)F_{a-h}(\theta)}{2h}.$$

In particular,

$$\widehat{\psi}(0) = 2a, \qquad |\widehat{\psi}(\pi)| \le \frac{1}{2h},$$

the latter because $MF_M(\pi) \in \{0, 1\}$. Put

$$\phi(n) = \psi(n-c) \cdot \frac{1 - (-1)^{n-c}}{2}.$$

Then ϕ is bounded above by the indicator counted by $N_{\mathrm{odd}}(a+h)$, while

$$\Phi(0) = \frac{\widehat{\psi}(0) - \widehat{\psi}(\pi)}{2} \ge a - \frac{1}{4h}.$$

Using the nonnegative majorant

$$\widetilde{\psi}(\theta) = \frac{(a+h)F_{a+h}(\theta) + (a-h)F_{a-h}(\theta)}{2h}$$

we have $|\Phi(j)| \le \frac{1}{2}\big(\widetilde{\psi}(2\pi j/v) + \widetilde{\psi}(2\pi j/v + \pi)\big)$, and Fejér aliasing gives

$$\sum_{j=1}^{v-1} |\Phi(j)| \le \frac{av}{h}.$$

Lemma 3.1 now yields the claimed bound. □

5. Exact-cardinality thinning

Fix a large integer k. Choose a prime q with

$$4k < q \le 8k,$$

which exists by Bertrand's postulate. Define

$$N = v(q+1), \qquad \Delta = \sqrt{(q+1)\log\big(64N\sqrt{q+1}\big)} + \frac{\pi}{2}, \qquad \Delta' = \Delta + \sqrt{q},$$
$$r = \lceil 15\Delta' \rceil, \qquad \Lambda = \Delta' + r,$$

and

$$h_0 = \Big\lceil \frac{v}{5(q+1)}\Big(r + 5\sqrt{q} + \frac{1}{4}\Big)\Big\rceil, \qquad \rho_0 = 6h_0, \qquad \tau_0 = \frac{1}{5}\sqrt{\frac{q+1}{vL\Lambda}}.$$

For all sufficiently large q these parameters satisfy

$$\rho_0 \le \frac{3L}{10}, \qquad 3\rho_0\tau_0 \le \frac{2}{5}, \qquad \tau_0 \le 0.05, \tag{5.1}$$

and

$$\Delta' = O\big(\sqrt{q\log q}\big), \qquad \Lambda = O\big(\sqrt{q\log q}\big), \qquad 0.09\Delta' \ge \sqrt{0.15(q+2)}. \tag{5.2}$$

Indeed $N \asymp q^3$ gives $\Delta \asymp \sqrt{q\log q}$, hence $r \asymp \Lambda \asymp \sqrt{q\log q}$ and $h_0 \asymp q^{3/2}\sqrt{\log q}$, while $L \asymp q^2$; since $(q+1)/(vL) \asymp q^{-3}$ we get $\tau_0 \asymp q^{-3/2}\Lambda^{-1/2} \asymp q^{-7/4}(\log q)^{-1/4}$, so $\rho_0\tau_0 \asymp q^{-1/4}(\log q)^{1/4} \to 0$. The last inequality in (5.2) holds as soon as $\log q \ge 7$, because $\log(64N\sqrt{q+1}) \ge 3.5\log q$ gives $\Delta' \ge \sqrt{3.5(q+1)\log q}$.

By Lemma 4.1 with $a = 5h_0$ and $h = h_0$, there are at least

$$\frac{q+1}{v}\Big(5h_0 - \frac{1}{4h_0}\Big) - 5\sqrt{q} \;\ge\; r$$

elements $d \in D$ satisfying

$$|d-c| \le \rho_0, \qquad d-c \text{ odd}.$$

Fix a set P of exactly r such elements. Since $\rho_0 \le 3L/10$,

$$w(d) \ge 0.7 \qquad (d \in P).$$

Put

$$Q_P = \sum_{d\in P} w(d), \qquad M = \sum_{d\in D\setminus P} w(d) = \kappa + R(0) - Q_P, \qquad \theta = \frac{k}{M},$$

where $|R(0)| \le \sqrt{q}$ by Lemma 3.2 at $y = 0$. Since $0 \le Q_P \le r$, we have $\kappa - \sqrt{q} - r \le M \le \kappa + \sqrt{q}$, and for sufficiently large k the inequalities $4k < q \le 8k$, $(q+1)/2 \le \kappa \le (q+2)/2$ and $r = o(k)$ imply

$$\frac{1}{5} \le \theta \le 1. \tag{5.3}$$

For $d \in D \setminus P$, independently choose $\xi_d \in \{0,1\}$ with

$$\mathbb{P}(\xi_d = 1) = p_d = \theta w(d),$$

and put $\xi_d = 0$ on P. Define

$$S = \{d-c : \xi_d = 1\}, \qquad T(y) = \sum_{s\in S} e^{isy}.$$

Then

$$\mathbb{E}|S| = \sum_{d\notin P} \theta w(d) = k.$$

Lemma 5.1 (Uniform noise bound)**.** *Let*

$$\mathcal{D}(y) = T(y) - \mathbb{E}T(y).$$

Then

$$\mathbb{P}\Big(\sup_{y\in\mathbb{R}} |\mathcal{D}(y)| > \Delta\Big) \le \frac{1}{16\sqrt{q+1}}.$$

Proof. For fixed y, Hoeffding's inequality applied separately to the real and imaginary parts gives

$$\mathbb{P}\big(|\mathcal{D}(y)| \ge t\big) \le 4\exp\Big(-\frac{t^2}{q+1}\Big).$$

Take the grid $y_\ell = 2\pi\ell/N$, $0 \le \ell < N$, and

$$t_0 = \sqrt{(q+1)\log\big(64N\sqrt{q+1}\big)}.$$

A union bound gives grid failure probability at most $1/(16\sqrt{q+1})$. Since $\mathcal{D}$ has degree at most c and $\|\mathcal{D}\|_\infty \le q+1$, Bernstein's inequality gives $\|\mathcal{D}'\|_\infty \le c(q+1)$. Every y lies within π/N of a grid point, hence

$$|\mathcal{D}(y) - \mathcal{D}(y_\ell)| \le \frac{\pi}{N}c(q+1) = \frac{\pi c}{v} < \frac{\pi}{2}.$$

Thus the grid bound t_0 implies the uniform bound $\Delta = t_0 + \pi/2$. □

Lemma 5.2 (Exact cardinality)**.** *Let X be a sum of s independent Bernoulli variables and suppose $\mathbb{E}X = k \in \mathbb{Z}$. Then*

$$\mathbb{P}(X = k) \ge \frac{1}{4\sqrt{s}}.$$

Proof. By Darroch's theorem [5] on the mode of a Poisson-binomial distribution, the integer mean k is a mode of X. Let $\sigma^2 = \mathrm{Var}(X) \le s/4$. If $\sigma = 0$, the assertion is immediate. Otherwise Chebyshev's inequality gives

$$\mathbb{P}(|X - k| < 2\sigma) \ge \frac{3}{4}.$$

The interval $(k - 2\sigma, k + 2\sigma)$ contains at most $4\sigma + 1$ integers, so one of them has probability at least

$$\frac{3}{4(4\sigma+1)} \ge \frac{1}{4\sqrt{s}}.$$

Since k is a mode, the same lower bound holds at k. □

Let G be the event $\sup_y |\mathcal{D}(y)| \le \Delta$. Here $s \le q+1$, and therefore Lemmas 5.1 and 5.2 give

$$\mathbb{P}(G^c) \le \frac{1}{16\sqrt{q+1}} < \frac{1}{4\sqrt{s}} \le \mathbb{P}(|S| = k).$$

Hence there exists a realization satisfying simultaneously

$$|S| = k, \qquad \sup_y |\mathcal{D}(y)| \le \Delta. \tag{5.4}$$

Fix such a realization from now on.

6. The three-zone estimate

By Lemma 3.2,

$$\mathbb{E}T(y) = \theta \sum_{d \in D \setminus P} w(d) e^{i(d-c)y} = \theta\big(m(y) + R(y) - Q(y)\big),$$

where

$$Q(y) = \sum_{d \in P} w(d) e^{i(d-c)y}.$$

Thus

$$T(y) = \theta m(y) + E(y), \qquad |E(y)| \le \Lambda, \tag{6.1}$$

because $|E| \le \theta\sqrt{q} + \theta Q_P + \Delta \le \sqrt{q} + r + \Delta = \Lambda$. Since $d - c$ is odd on P, for every real t,

$$T(\pi + 3t) = \theta\big(m(\pi + 3t) + R(\pi + 3t) + Q_0(t)\big) + \mathcal{D}(\pi + 3t), \tag{6.2}$$

where

$$Q_0(t) = \sum_{d \in P} w(d) e^{3i(d-c)t}.$$

Moreover, since $\rho(x) = \mathrm{Re}\big(e^{ix/2}T(3x)\big)$ and $g(x) = \mathrm{Re}\big(e^{ix}T(6x)\big)$,

$$\rho(x) \ge \cos(x/2)\theta m(3x) - \Lambda, \qquad g(x) \le \cos(x)\theta m(6x) + \Lambda. \tag{6.3}$$

Because f_A is even and 2π-periodic, it suffices to consider $x \in [0, \pi]$. Write $A = A(S)$ and $f_A = \rho^2 - k/2 - g/2$ (Proposition 2.1).

Zone I: $0 \le x \le \tau_0$**.** Put $\mu = \theta m(3x)$. By (3.2) and $\tau_0 \le 0.05$,

$$\theta m(6x) \le 1.024\mu, \qquad \cos(x/2) \ge 0.998.$$

Therefore

$$g(x) \le 1.024\mu + \Lambda, \qquad \rho(x) \ge 0.998\mu - \Lambda.$$

If $\mu \le 4\Lambda$, then $g(x) \le 5.1\Lambda$, and hence

$$f_A(x) \ge -\frac{k}{2} - 2.55\Lambda.$$

If $\mu > 4\Lambda$, then

$$\rho(x) \ge 0.748\mu,$$

so $\rho(x)^2 \ge 0.559\mu^2 \ge 0.637\mu \ge g(x)/2$ (as $\mu > 4\Lambda \ge 2$), and in particular

$$f_A(x) \ge -\frac{k}{2}.$$

Zone II: $x = \pi/3 + t$, $|t| \le \tau_0$. From (6.3), $\cos x \le 0.55$ and $m \le \kappa \le (q+2)/2$,

$$\frac{g(x)}{2} \le 0.15(q+2) + \frac{\Lambda}{2}.$$

Set $\phi = \pi/6 + t/2$. By (6.2),

$$\rho(x) = \theta m(\pi+3t)\cos\phi + \mathrm{Re}\Big(e^{i\phi}\big(\theta R(\pi+3t) + \mathcal{D}(\pi+3t)\big)\Big) + \theta\sum_{d\in P} w(d)\cos\big(\phi + 3(d-c)t\big).$$

The first term is nonnegative and the second is at least $-\Delta'$. For $d \in P$, (5.1) gives

$$|\phi + 3(d-c)t| \le \frac{\pi}{6} + \frac{1}{20} + 3\rho_0\tau_0 \le \frac{\pi}{6} + \frac{9}{20} < 0.974,$$

so

$$\cos\big(\phi + 3(d-c)t\big) \ge 0.56.$$

Using $w(d) \ge 0.7$, $\theta \ge 1/5$, and $r \ge 15\Delta'$ gives

$$\theta\sum_{d\in P} w(d)\cos\big(\phi + 3(d-c)t\big) \ge 0.2\cdot 0.7\cdot 0.56\cdot 15\Delta' \ge 1.17\Delta'.$$

Consequently

$$\rho(x) \ge 0.17\Delta' \ge 0.09\Delta' \ge \sqrt{0.15(q+2)}.$$

Thus $\rho(x)^2$ absorbs the potentially positive part of $g(x)/2$, and

$$f_A(x) \ge -\frac{k}{2} - \frac{\Lambda}{2}.$$

Zone III: the remainder of $[0,\pi]$. If $x \ge \pi/2$, then $\cos x \le 0$, so (6.3) gives $g(x) \le \Lambda$ and hence

$$f_A(x) \ge -\frac{k}{2} - \frac{\Lambda}{2}.$$

If $x < \pi/2$ but x lies outside Zones I and II, then $6x \in (6\tau_0, 3\pi)$ has distance at least $\delta := 6\tau_0$ from $2\pi\mathbb{Z}$, so $3x$ has distance at least $\delta/2$ from $\pi\mathbb{Z}$ and

$$|\sin 3x| \ge \sin(\delta/2) \ge 0.49\delta,$$

using $\delta \le 0.3$. By (3.1) and the definition of τ_0,

$$m(6x) \le \frac{q+1}{vL\sin^2(3x)} \le \frac{q+1}{0.24vL\delta^2} = \frac{25}{8.64}\cdot\Lambda \le 3\Lambda.$$

Since $\theta \le 1$,

$$g(x) \le 4\Lambda,$$

and therefore

$$f_A(x) \ge -\frac{k}{2} - 2\Lambda.$$

Combining the three zones gives

$$\min_x f_A(x) \geq -\frac{k}{2} - 2.55\Lambda = -\frac{k}{2} - O\big(\sqrt{k \log k}\big), \tag{6.4}$$

since $q \asymp k$ and $\Lambda = O(\sqrt{q \log q})$.

Theorem 6.1 (Base cardinalities)**.** *For every sufficiently large integer k there exists a set $A \subset \mathbb{N}$ with $|A| = k(k-1)$ such that*

$$\min_x f_A(x) \geq -\frac{k}{2} - O\big(\sqrt{k \log k}\big).$$

Equivalently, $\lambda(k(k-1)) \leq k/2 + O(\sqrt{k \log k})$.

This proves Theorem 1.2.

Remark 6.2. The constants in the definitions of Section 5 are chosen for simplicity of verification, not for size: (5.1) forces q to be of order at least 10^5, and at such sizes 2.55Λ exceeds $150\sqrt{k \log k}$. The bound is therefore asymptotic in nature. The values actually attained by the construction at moderate sizes are far better; see Section 10.

7. Passage to every cardinality

Lemma 7.1 (Subadditivity)**.** *For nonnegative integers a, b,*

$$\lambda(a+b) \leq \lambda(a) + \lambda(b).$$

Proof. The cases $a = 0$ or $b = 0$ are immediate. For $a, b \geq 1$, choose finite sets $A, B \subset \mathbb{N}$ of sizes a, b. Choose an integer $t > \max A$ sufficiently large that $A \cap tB = \emptyset$. Then

$$f_{A \cup tB}(x) = f_A(x) + f_B(tx),$$

so

$$\min_x f_{A \cup tB}(x) \geq \min_x f_A(x) + \min_y f_B(y).$$

Taking infima proves the claim. □

We use the explicit form of the Sidon-difference estimate [10, Fact 2.1],

$$\lambda(m) \leq \sqrt{2m} + \frac{1}{2} \qquad (m \geq 1). \tag{7.1}$$

Theorem 7.2 (Half-constant upper bound)**.** *As $n \to \infty$,*

$$\lambda(n) \leq \frac{1}{2}\sqrt{n} + O\big(n^{1/4}\sqrt{\log n}\big).$$

Consequently,

$$\limsup_{n \to \infty} \frac{\lambda(n)}{\sqrt{n}} \leq \frac{1}{2}.$$

Proof. Let k be maximal with $k(k-1) \le n$ and put

$$m = n - k(k-1).$$

Then $0 \le m < 2k$ and $k = \sqrt{n} + O(1)$. By Theorem 6.1, Lemma 7.1, and (7.1),

$$\begin{aligned}\lambda(n) &\le \lambda(k(k-1)) + \lambda(m)\\ &\le \frac{k}{2} + O\big(\sqrt{k\log k}\big) + O(\sqrt{m})\\ &= \frac{1}{2}\sqrt{n} + O\big(n^{1/4}\sqrt{\log n}\big).\end{aligned}$$

Dividing by $\sqrt{n}$ and letting $n \to \infty$ gives the limsup statement. □

This proves Theorem 1.3.

8. Limits of the method

All known constructions, including the one above, are of the form "square of a trigonometric polynomial, minus the diagonal, minus a correction that is handled analytically". We record an exact identity that shows what such a construction must achieve in order to have leading constant below $1/2$. For a trigonometric polynomial $F(x) = \sum_\gamma c_\gamma e^{i\gamma x}$ with real (not necessarily integer) frequencies γ, write $\mathrm{M}(F) = c_0$ for its mean; thus $\mathrm{M}(F) = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} F$.

Proposition 8.1 (L^4 criterion)**.** *Let $A \subset \mathbb{N}$ be finite with $|A| = n$, and suppose that for some $\lambda \ge 0$, some trigonometric polynomial P with complex coefficients and real frequencies, and some real trigonometric polynomial h with $\mathrm{M}(h) = 0$ whose frequency support is disjoint from $A \cup (-A)$,*

$$\lambda + f_A(x) = |P(x)|^2 - h(x) \qquad (x \in \mathbb{R}).$$

Then $\mathrm{M}(|P|^2) = \lambda$, $\mathrm{M}(|P|^4) = \lambda^2 + \frac{n}{2} + \mathrm{M}(h^2)$, and

$$\frac{\lambda}{\sqrt{n}} = \frac{\mathrm{M}(|P|^2)}{\sqrt{2\big(\mathrm{M}(|P|^4) - \mathrm{M}(|P|^2)^2 - \mathrm{M}(h^2)\big)}}.$$

In particular $\lambda < \frac{1}{2}\sqrt{n}$ if and only if

$$\mathrm{M}(|P|^4) - \mathrm{M}(h^2) > 3\,\mathrm{M}(|P|^2)^2.$$

Proof. Taking means, $\mathrm{M}(f_A) = 0$ and $\mathrm{M}(h) = 0$ give $\mathrm{M}(|P|^2) = \lambda$. Squaring, $\mathrm{M}(|P|^4) = \mathrm{M}\big((\lambda + f_A + h)^2\big) = \lambda^2 + \mathrm{M}(f_A^2) + \mathrm{M}(h^2)$, because $\mathrm{M}(f_A) = \mathrm{M}(h) = 0$ and $\mathrm{M}(f_A h) = 0$ by the disjointness of supports. Finally $\mathrm{M}(f_A^2) = n/2$, since the n frequencies of f_A are distinct positive integers. The displayed formula and the equivalence follow. □

In Proposition 2.1 one has $P = \rho$, $\lambda = k/2$, $h = g/2$, $\mathrm{M}(g^2) = k/2$, and $\mathrm{M}(\rho^4) = \frac{3}{4}k^2 - \frac{3}{8}k$, so the criterion reads $\frac{3}{4}k^2 - \frac{1}{2}k > \frac{3}{4}k^2$, which fails; this is consistent with $\lambda/\sqrt{n} = \frac{1}{2}(1 - 1/k)^{-1/2}$.

For a finite $B \subset \mathbb{R}$ and $\gamma \in \mathbb{R}$ write $r_B(\gamma) = \#\{(\beta, \beta') \in B^2 : \beta - \beta' = \gamma\}$. When P has coefficients of constant modulus, the criterion takes a sharper, pointwise form.

Proposition 8.2 (Multiplicity bound)**.** *Let $A \subset \mathbb{N}$ be finite with $|A| = n$, and suppose that $\lambda + f_A = |P|^2 - h$ with h as in Proposition 8.1 and*

$$P(x) = c \sum_{\beta \in B} \varepsilon_\beta e^{i\beta x}, \qquad c > 0, \quad |\varepsilon_\beta| = 1,$$

for a finite set $B \subset \mathbb{R}$. Put $m = 1/(2c^2)$. Then for every $\gamma \in A \cup (-A)$,

$$\sum_{\beta - \beta' = \gamma} \varepsilon_\beta \overline{\varepsilon_{\beta'}} = m, \qquad \textit{so in particular} \qquad r_B(\gamma) \geq m,$$

and

$$\lambda \;\geq\; \sqrt{\frac{n}{2m(1 - 1/|B|)}} \;>\; \sqrt{\frac{n}{2m}}.$$

Consequently $\lambda < \frac{1}{2}\sqrt{n}$ forces $m > 2$: every frequency of A then has at least three representations as a difference of elements of B.

Proof. Taking means in $\lambda + f_A = |P|^2 - h$ gives $\lambda = \mathrm{M}(|P|^2) = c^2|B|$, because $\mathrm{M}(f_A) = \mathrm{M}(h) = 0$. The coefficient of $e^{i\gamma x}$ in $|P|^2$ is $c^2 \sum_{\beta - \beta' = \gamma} \varepsilon_\beta \overline{\varepsilon_{\beta'}}$. For $\gamma \in A \cup (-A)$ the coefficient of $e^{i\gamma x}$ in $\lambda + f_A$ is $\frac{1}{2}$ and that in h is 0, so the sum equals $1/(2c^2) = m$, and $m \leq r_B(\gamma)$ by the triangle inequality. Summing over the $2n$ elements of $A \cup (-A)$, which are distinct and nonzero,

$$2nm \;\leq \sum_{\gamma \in A \cup (-A)} r_B(\gamma) \;\leq \sum_{\gamma \neq 0} r_B(\gamma) = |B|^2 - |B|.$$

Since $|B| = \lambda/c^2 = 2m\lambda$, the right-hand side equals $4m^2\lambda^2(1 - 1/|B|)$, and the inequality reads $n \leq 2m\lambda^2(1 - 1/|B|)$. If $\lambda < \frac{1}{2}\sqrt{n}$, then $n/4 > \lambda^2 \geq n/(2m(1 - 1/|B|))$ gives $m > 2/(1 - 1/|B|) > 2$, and $r_B(\gamma) \geq m > 2$ for every $\gamma \in A \cup (-A)$. □

The proposition uses only the mean of $|P|^2$ and its coefficients on $A \cup (-A)$; it places no condition on h beyond the support condition of Proposition 8.1. For the identity of Proposition 2.1 one has $P = \rho = \frac{1}{2}\sum_{\beta \in B} e^{i\beta x}$ with $B = \{\pm\alpha_j\}$, so $c = \frac{1}{2}$, $m = 2$, $|B| = 2k$, and every frequency of $A(S)$ has $r_B = 2$ exactly; the bound reads $\frac{k}{2} \geq \frac{1}{2}\sqrt{k(k-1) \cdot 2k/(2k-1)}$, which is $\lambda/\sqrt{n} = \frac{1}{2}(1 - 1/k)^{-1/2}$ in another form. The multiplicity two is supplied by the symmetry $\beta \mapsto -\beta$.

Lemma 8.3 (Multiplicities of symmetric supports)**.** *Let $B \subset \mathbb{R} \setminus \{0\}$ be finite with $B = -B$. Then for every $\gamma \neq 0$,*

$$r_B(\gamma) \equiv \mathbf{1}[\gamma/2 \in B] \pmod 2.$$

Proof. The map $(\beta, \beta') \mapsto (-\beta', -\beta)$ is an involution on the set of representations of γ. Its fixed points are the pairs $(\beta, -\beta)$, which exist if and only if $\gamma = 2\beta$ with $\beta \in B$, and then there is exactly one. □

For $B = \{\pm\alpha_j\}$ with $\alpha_j = 3s_j + \frac{1}{2}$ and S Sidon, Lemma 8.3 says that every frequency $\alpha_i \pm \alpha_j$ $(i \neq j)$ has multiplicity exactly two while the frequencies $2\alpha_j = 6s_j + 1$ have multiplicity one; the latter are precisely the frequencies of the correction g in Proposition 2.1. Thus the identity of Section 2 is the extremal case of Lemma 8.3.

Remark 8.4. Proposition 8.2 bounds the constant λ of the representation, whereas the quantity of interest is

$$-\min_x f_A(x) = \lambda + \sup_x \big(h(x) - |P(x)|^2\big).$$

Within the framework of Proposition 8.1 a leading constant below $\frac{1}{2}$ can therefore arise in only two ways: either every frequency of A has multiplicity $m > 2$ in B in the sense of Proposition 8.2, or $h \leq |P|^2 - \varepsilon\lambda$ on all of $\mathbb{R}$ for some fixed $\varepsilon > 0$, that is, the correction is negative by a fixed proportion of λ at every zero of P. We know of no support B for which either can be carried out. For the first route, $\beta \mapsto -\beta$ is the only nontrivial automorphism of $\mathbb{Z}$ and supplies exactly multiplicity two (Lemma 8.3), so larger multiplicities must come from additive coincidences in B; in the examples we have examined these are accompanied by a comparable number of frequencies whose coefficient in $|P|^2$ is not 1, and the resulting h is of order λ rather than $o(\lambda)$. For a union of translates $B = C + D$ with C Sidon, the multiplicities follow the profile of r_D, which is never constant on $D - D$; for the lift to $\mathbb{Z}$ of a cyclic difference set with parameter $\lambda_0 \geq 2$, the λ_0 representations of each nonzero residue are distributed over two integers. For the second route, in the construction of Section 6 the values of g at the zeros of ρ are essentially those of the random part $\mathcal{D}$ (Remark 8.5), and nothing forces them to be negative. Lowering the constant below $\frac{1}{2}$ therefore appears to require an idea outside the square-identity paradigm.

Remark 8.5 (Sharpness for the present construction)**.** For the sets $A = A(S)$ of Theorem 6.1, at every zero x_0 of ρ one has $f_A(x_0) = -k/2 - g(x_0)/2$, so

$$-\min_x f_A(x) \;\geq\; \frac{k}{2} + \frac{1}{2}\sup\{g(x_0) : \rho(x_0) = 0\}.$$

Thus the leading term $k/2$ could only be improved if g were negative at every zero of ρ. Away from the neighbourhoods of 0, $\pi/3$, $2\pi/3$ and π both ρ and g consist of the random part $\mathcal{D}$ evaluated at $3x$ and $6x$ respectively, whose values at a fixed point are essentially uncorrelated, and ρ has zeros on the scale $1/v$; one therefore expects the supremum above to be positive and of order $\sqrt{k \log k}$, the uniform size of $\mathcal{D}$, so that both the leading term and the order of the error term in Theorem 6.1 should be sharp for this construction up to the logarithm. We do not prove this. The numerical minima in Section 10 are attained at points where $\rho^2 < 1$, with $-g/2$ between $1.5\sqrt{k}$ and $1.8\sqrt{k}$.

9. Lower bounds: dense sets and the L^1 norm

The sets of Theorem 6.1 have all their frequencies below $3v \le 200n$ (Section 5: $S \subset [-c, c]$ with $c = (v-1)/2$, so every frequency $3|s_i - s_j|$ and $|3(s_i + s_j) + 1|$ is at most $3(v-1)+1$; and $q \le 8k$, $v = q^2 + q + 1$, $n = k(k-1)$, $k \ge 29$ give $3v \le 200n$). For sets of this kind a polynomial lower bound follows from Sanders' work on the analogous problem in finite abelian groups [13], where for a symmetric $A \subset G$ one considers $M_G(A) = \sup_\gamma(-\widehat{\mathbf{1}_A}(\gamma))$, the supremum over the characters of G. Sanders proves that a symmetric set $A \ni 0$ of density α in G either satisfies $|A \triangle H| \le K$ for some subgroup $H \le G$ or has $M_G(A) \ge K^{\alpha/5}/64$, for every $0 < K \le |A|/2^{13}$ [13, Theorem 4.1], and that a symmetric $A \subset \mathbb{Z}/p\mathbb{Z}$ of size $(p+1)/2$ has $M_{\mathbb{Z}/p\mathbb{Z}}(A) \ge cp^{c'}$ with absolute constants $c, c' > 0$, where he indicates that $c' = 1/3$ can be arranged [13, Theorem 1.2]. If $A \subset [1, Cn]$ has n elements, then $A \cup (-A) \cup \{0\}$ embeds in $\mathbb{Z}/p\mathbb{Z}$, for a prime $p \in (4Cn, 8Cn]$, as a symmetric set of density $\alpha > 1/(4C)$ whose symmetric difference with each of the two subgroups of $\mathbb{Z}/p\mathbb{Z}$ has at least $2n$ elements, and the supremum over the characters of $\mathbb{Z}/p\mathbb{Z}$ is at most $-2\min f_A - 1$; hence $-\min f_A \ge cn^{\alpha/5}$ with an absolute $c > 0$, an exponent that tends to zero as C grows. Proposition 9.7 below gives the exponent $1/4$ for every C, only the constant depending on C, with an elementary selection of the auxiliary shift. In the one case where Sanders' bound has an absolute exponent, sets of density exactly $1/2$ in $\mathbb{Z}/p\mathbb{Z}$, his exponent $1/3$ is the better one.

The tool is the boundary estimate of Bedert [1, Proposition 7.3]. In Section 9.1 we reproduce its proof with the L^1 norm of the cosine sum carried through, which replaces one factor of $-\min f_A$ by $\|f_A\|_1$ and makes the constant explicit. Section 9.2 treats sets with frequencies $O(n)$, and Section 9.3 combines the estimate with Shankar's total-boundary identity

[14] to obtain a lower bound for general sets in which two factors of $-\min f_A$ are replaced by $\|f_A\|_1$.

Notation. Throughout this section $A' \subset \mathbb{Z} \setminus \{0\}$ is a finite symmetric set, $N = |A'|$, $a_m = \mathbf{1}_{A'}(m)$, $e(y) = e^{2\pi i y}$, and

$$F(y) = \widehat{\mathbf{1}_{A'}}(y) := \sum_{a \in A'} e(ay),$$

a real even trigonometric polynomial on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. More generally $\widehat{\mathbf{1}_X}(y) = \sum_{x \in X} e(xy)$ for finite $X \subset \mathbb{Z}$. The Fourier coefficients of a function g on $\mathbb{T}$ are $\widehat{g}(m) = \int_{\mathbb{T}} g(y) e(-my)\, dy$, so that $\widehat{\widehat{\mathbf{1}_X}} = \mathbf{1}_X$. All integrals are over $\mathbb{T}$ with respect to the probability Haar measure, $\|g\|_r = (\int |g|^r)^{1/r}$, and $(g * h)(y) = \int g(y-z) h(z)\, dz$, so that $\widehat{g * h} = \widehat{g}\widehat{h}$ and in particular $\widehat{\mathbf{1}_X} * \widehat{\mathbf{1}_Y} = \widehat{\mathbf{1}_{X \cap Y}}$. We have $\int F = 0$ and $\int F^2 = N$. We fix $K > 0$ with

$$F + K \geq 0 \quad \text{on } \mathbb{T},$$

and write $u = F + K \geq 0$, so that $\int u = K$ and $\widehat{u}(m) = a_m$ for $m \neq 0$; hence $u * u = F + K^2 = u + K^2 - K$. We put $\mu = \|F\|_1$. Since $\int F = 0$ we have $\int F_+ = \int F_- = \mu/2$, where $F_\pm = \max(\pm F, 0)$, and since $F_- \leq K$,

$$\mu \leq 2K, \qquad \|F_-\|_2^2 \leq K\mu/2, \qquad \int F_-^j \leq K^{j-1}\mu/2 \quad (j \geq 1). \tag{9.1}$$

Moreover $K \geq 1$ when $A' \neq \emptyset$: for $a_0 \in A'$, $1 = \int u(y) e(-a_0 y)\, dy \leq \int u = K$. For $t \in \mathbb{Z} \setminus \{0\}$ let

$$B_t(A') := \{a \in A' : a - t \in A',\ a + t \notin A'\},$$

the set of right endpoints of the maximal arithmetic progressions with common difference t and at least two terms contained in A'.

For $A \subset \mathbb{N}$ finite we write $\lambda_A := -\min_x f_A(x)$ and $\|f_A\|_1 := \frac{1}{2\pi}\int_0^{2\pi} |f_A|$. With $A' = A \cup (-A)$ one has $F(y) = 2 f_A(2\pi y)$, so the hypotheses above hold with $N = 2n$, $K = 2\lambda_A$ and $\mu = 2\|f_A\|_1$, and (9.1) reads $\|f_A\|_1 \leq 2\lambda_A$.

9.1. **The boundary estimate.** Bedert [1, Proposition 7.3] proves that $|B_t(A')| \ll K^4$ for every $t \neq 0$. His argument gives slightly more: one factor K can be replaced by $\mu = \|F\|_1$, and the constant can be made explicit. Since this refinement is what Theorem 9.11 uses, we give the argument in full. Apart from the bookkeeping of L^1 norms it is Bedert's; the only changes are that $\|f_t\|_1 \leq 4\mu$ replaces $\|f_t\|_1 \leq 8K$ in the bound for the function $f_t * f_t$ below, and that the L^1 norms of the positive and negative parts of F are kept in his decomposition lemma.

Proposition 9.1 (Boundary estimate)**.** *In the above setting, for every $t \in \mathbb{Z} \setminus \{0\}$,*

$$|B_t(A')| \;\le\; C_1 K^3 \mu \;\le\; 2C_1 K^4, \qquad C_1 = 2 \cdot 10^5.$$

We need three lemmas. The first is [1, Lemma 3.3].

Lemma 9.2. *Let g, h be real integrable functions on $\mathbb{T}$ with $\int g = \int h = 0$, $g \ge -G$ and $h \ge -H$ for constants $G, H \ge 0$. Then $g * h \ge -\frac{1}{2}\big(G\|h\|_1 + H\|g\|_1\big)$.*

Proof. Write $g = g_+ - g_-$ and $h = h_+ - h_-$ with $0 \le g_- \le G$, $0 \le h_- \le H$, $\int g_+ = \|g\|_1/2$ and $\int h_+ = \|h\|_1/2$. Then $g * h \ge -(g_- * h_+) - (g_+ * h_-) \ge -G \int h_+ - H \int g_+$. □

For the second lemma write $T_1 = F_+$ and $T_2 = -F_-$, so that $F = T_1 + T_2$, $0 \le T_1 \le u$, $-K \le T_2 \le 0$, $\|T_1\|_1 = \|T_2\|_1 = \mu/2$ and $\|T_2\|_2^2 \le K\mu/2$, and put $e_s(y) = e(sy)$ for $s \in \mathbb{Z}$. Besides Parseval and Young's inequality $\|g * h\|_2 \le \|g\|_1 \|h\|_2$ we use the following splitting device: if g is a function on $\mathbb{T}$ with $|g| \le u * u = u + K^2 - K$, then $g = g' + g''$ with $g' = g \min(1, u/|g|)$ (so $g' = g$ where $|g| \le u$), and

$$|g'| \le u, \qquad |g''| = (|g| - u)_+ \le K^2, \qquad |g''| \le |g|, \tag{9.2}$$

hence $\|g''\|_2 \le (\|g''\|_\infty \|g''\|_1)^{1/2} \le K\|g\|_1^{1/2}$.

Lemma 9.3 (Decomposition)**.** *For every $t \ne 0$ there are functions Q_1, Q_2 on $\mathbb{T}$ such that*

$$\widehat{\mathbf{1}_{B_t(A')}} = Q_1 + Q_2, \qquad |Q_1| \le 3u, \qquad Q_1(-y) = \overline{Q_1(y)},$$
$$\|Q_2\|_2 \le 6K\mu + \sqrt{2}\, K^{3/2}\mu^{1/2} \le 7\sqrt{2}\, K^{3/2}\mu^{1/2}.$$

Proof. Write $B = B_t(A')$ and $A'_t = A' \cap (A' + t)$, so that $A'_{-t} = -A'_t$, $B = A'_t \setminus A'_{-t}$ and $-B = A'_{-t} \setminus A'_t$. Since $(e_s F)^\wedge(m) = a_{m-s}$, comparing Fourier coefficients gives

$$\widehat{\mathbf{1}_B} = \tfrac{1}{2}\Big(\widehat{\mathbf{1}_{A'_t}} + \widehat{\mathbf{1}_{A'_{-t}}} - 2\widehat{\mathbf{1}_{A'_t \cap A'_{-t}}}\Big) + \tfrac{1}{2} F * \big((e_t - e_{-t})F\big), \tag{9.3}$$

the first bracket being $\widehat{\mathbf{1}_B} + \widehat{\mathbf{1}_{-B}}$ and the last term $\widehat{\mathbf{1}_B} - \widehat{\mathbf{1}_{-B}}$ (its mth coefficient is $a_m(a_{m-t} - a_{m+t})$), and also $\widehat{\mathbf{1}_{A'_t}} = F * (e_t F)$ and $\widehat{\mathbf{1}_{A'_t \cap A'_{-t}}} = \widehat{\mathbf{1}_{A'_t}} * (e_{-t} F)$. We decompose each term.

The sets $A'_{\pm t}$. We have $F * (e_t F) = T_1 * (e_t T_1) + T_1 * (e_t T_2) + T_2 * (e_t T_1) + T_2 * (e_t T_2)$. The first term has modulus at most $T_1 * T_1 \le u * u$, so (9.2) writes it as $R'_1 + V$ with $|R'_1| \le u$ and $\|V\|_2 \le K\|T_1 * (e_t T_1)\|_1^{1/2} \le K\|T_1\|_1 = K\mu/2$. Each of the other three terms has L^2 norm at most $\|T_1\|_1 \|T_2\|_2$ or $\|T_2\|_1 \|T_2\|_2$

by Young's inequality, that is, at most $\frac{\mu}{2}(\frac{K\mu}{2})^{1/2} = \frac{1}{2\sqrt{2}}K^{1/2}\mu^{3/2} \le \frac{1}{2}K\mu$, where we used $\mu \le 2K$. Hence $\widehat{\mathbf{1}_{A'_t}} = R'_1 + R'_2$ with

$$|R'_1| \le u, \qquad \|R'_1\|_1 \le K, \qquad \|R'_2\|_2 \le 2K\mu,$$

and in the same way $\widehat{\mathbf{1}_{A'_{-t}}} = R''_1 + R''_2$ with the same bounds.

The set $A'_t \cap A'_{-t}$. We have $\widehat{\mathbf{1}_{A'_t\cap A'_{-t}}} = R'_1 * (e_{-t}T_1) + R'_1 * (e_{-t}T_2) + R'_2 * (e_{-t}F)$. The first term has modulus at most $u * u$ and (9.2) writes it as $R'''_1 + v$ with $|R'''_1| \le u$ and $\|v\|_2 \le K\big(\|R'_1\|_1\|T_1\|_1\big)^{1/2} \le K^{3/2}(\mu/2)^{1/2}$; the second has L^2 norm at most $\|R'_1\|_1\|T_2\|_2 \le K^{3/2}(\mu/2)^{1/2}$; the third has Fourier coefficients $\widehat{R'_2}(m)a_{m+t}$, so its L^2 norm is at most $\|R'_2\|_2 \le 2K\mu$. Thus $\widehat{\mathbf{1}_{A'_t\cap A'_{-t}}} = R'''_1 + R'''_2$ with

$$|R'''_1| \le u, \qquad \|R'''_2\|_2 \le \sqrt{2}\,K^{3/2}\mu^{1/2} + 2K\mu.$$

The odd part. Write $(e_t - e_{-t})F = S'_1 + S'_2$ with $S'_1 = (e_t - e_{-t})T_1$ and $S'_2 = (e_t - e_{-t})T_2$; then $|S'_1| \le 2T_1$, $\|S'_1\|_1 \le \mu$, $\|S'_2\|_1 \le \mu$ and $\|S'_2\|_2 \le 2\|T_2\|_2 \le (2K\mu)^{1/2}$. Now $F*(S'_1+S'_2) = T_1*S'_1 + T_1*S'_2 + T_2*S'_1 + T_2*S'_2$. The function $\frac{1}{2}T_1*S'_1$ has modulus at most $T_1*T_1 \le u*u$, so by (9.2) the first term equals S_1+V_2 with $|S_1| \le 2u$ and $\|V_2\|_2 \le 2K\big(\frac{1}{2}\|T_1\|_1\|S'_1\|_1\big)^{1/2} \le K\mu$. The other three terms have L^2 norms at most $\|T_1\|_1\|S'_2\|_2 \le \frac{\mu}{2}(2K\mu)^{1/2}$, $\|S'_1\|_1\|T_2\|_2 \le \mu(K\mu/2)^{1/2}$ and $\|S'_2\|_1\|T_2\|_2 \le \mu(K\mu/2)^{1/2}$, each equal to $K^{1/2}\mu^{3/2}/\sqrt{2} \le K\mu$. Hence $F * \big((e_t - e_{-t})F\big) = S_1 + S_2$ with

$$|S_1| \le 2u, \qquad \|S_2\|_2 \le 4K\mu.$$

Conclusion. Put $Q^0_1 = \frac{1}{2}(R'_1 + R''_1 - 2R'''_1) + \frac{1}{2}S_1$ and $Q^0_2 = \frac{1}{2}(R'_2 + R''_2 - 2R'''_2) + \frac{1}{2}S_2$. By (9.3), $\widehat{\mathbf{1}_B} = Q^0_1 + Q^0_2$, $|Q^0_1| \le \frac{1}{2}(u+u+2u) + u = 3u$, and

$$\|Q^0_2\|_2 \le \tfrac{1}{2}\big(2K\mu+2K\mu+2\sqrt{2}\,K^{3/2}\mu^{1/2}+4K\mu\big)+2K\mu = 6K\mu+\sqrt{2}\,K^{3/2}\mu^{1/2}.$$

Finally, $\widehat{\mathbf{1}_B}(-y) = \overline{\widehat{\mathbf{1}_B}(y)}$ and u is even, so $Q_j(y) := \frac{1}{2}\big(Q^0_j(y) + \overline{Q^0_j(-y)}\big)$ $(j = 1, 2)$ satisfy $Q_1 + Q_2 = \widehat{\mathbf{1}_B}$, $|Q_1| \le 3u$, $Q_1(-y) = \overline{Q_1(y)}$ and $\|Q_2\|_2 \le \|Q^0_2\|_2$. The last inequality of the lemma is $6K\mu \le 6\sqrt{2}K^{3/2}\mu^{1/2}$, that is, $\mu \le 2K$. □

The third lemma concerns Bedert's twisted function

$$f_t(y) = 2\big(1 + \sin 2\pi ty\big)F(y).$$

Since $(1+\sin 2\pi ty)(F+K) \ge 0$ and $0 \le 1+\sin 2\pi ty \le 2$, we have $f_t \ge -4K$ and $\|f_t\|_1 \le 4\mu$; and since $2\sin 2\pi ty = -i(e_t - e_{-t})$,

$$\widehat{f_t}(m) = 2a_m - ia_{m-t} + ia_{m+t} \in \{0,\ 2,\ 2 \pm i,\ \pm i\}, \tag{9.4}$$

which vanishes at $m = 0$ by symmetry, so $\int f_t = 0$, and equals $2 - i$ for $m \in B_t(A')$. By Lemma 9.2, $f_t * f_t \geq -\frac{1}{2}(4K \cdot 4\mu + 4K \cdot 4\mu) = -16K\mu$. Put

$$r_t(y) = \big(1 - \cos(2\pi t y + \pi/4)\big)(f_t * f_t)(y) \;\geq\; -32K\mu,$$

and let $\psi_t(y) = \frac{1}{2}(r_t(y) + r_t(-y))$ and $\varphi_t(y) = \frac{1}{2}(r_t(y) - r_t(-y))$ be its even and odd parts. Since $\psi_t \pm \varphi_t \geq -32K\mu$,

$$|\varphi_t(y)| \leq \psi_t(y) + 32K\mu \qquad (y \in \mathbb{T}). \tag{9.5}$$

Write $\rho_m = \widehat{r_t}(m)$. As r_t is real, $\rho_{-m} = \overline{\rho_m}$, so $\widehat{\psi_t}(m) = \operatorname{Re}\rho_m$ and $\widehat{\varphi_t}(m) = i \operatorname{Im}\rho_m$. Since $\cos(2\pi t y + \pi/4) = \frac{1}{2}(e^{i\pi/4}e_t + e^{-i\pi/4}e_{-t})$ and $(f_t * f_t)^\wedge = \widehat{f_t}^{\,2}$,

$$\rho_m = \widehat{f_t}(m)^2 - \frac{1+i}{2\sqrt{2}}\widehat{f_t}(m-t)^2 - \frac{1-i}{2\sqrt{2}}\widehat{f_t}(m+t)^2. \tag{9.6}$$

Lemma 9.4 (Coefficient bounds)**.** *For all $m \in \mathbb{Z}$, $\operatorname{Re}\rho_m \leq 4 + \frac{1}{\sqrt{2}}$ and $|\operatorname{Im}\rho_m| \leq 9$. For $m \in B_t(A')$, $-\operatorname{Im}\rho_m \geq 4 + \sqrt{2}$.*

Proof. By (9.4) each square in (9.6) lies in $Z = \{0, 4, 3+4i, 3-4i, -1\}$. For $z \in Z$ one has $\operatorname{Re} z \leq 4$, $-\operatorname{Re}\frac{(1+i)z}{2\sqrt{2}} = \frac{\operatorname{Im} z - \operatorname{Re} z}{2\sqrt{2}} \leq \frac{1}{2\sqrt{2}}$ and $-\operatorname{Re}\frac{(1-i)z}{2\sqrt{2}} = -\frac{\operatorname{Re} z + \operatorname{Im} z}{2\sqrt{2}} \leq \frac{1}{2\sqrt{2}}$, which gives the first bound; and $|\operatorname{Im} z| \leq 4$, $\big|\operatorname{Im}\frac{(1\pm i)z}{2\sqrt{2}}\big| \leq \frac{7}{2\sqrt{2}}$, which gives $|\operatorname{Im}\rho_m| \leq 4 + \frac{7}{\sqrt{2}} < 9$. If $m \in B_t(A')$ then $a_m = a_{m-t} = 1$ and $a_{m+t} = 0$, so $\widehat{f_t}(m)^2 = (2-i)^2 = 3 - 4i$, $\widehat{f_t}(m-t) = 2 + i(1 - a_{m-2t}) \in \{2, 2+i\}$ and $\widehat{f_t}(m+t) = -i(1 - a_{m+2t}) \in \{-i, 0\}$. Substituting the four possibilities into (9.6) gives

$$-\operatorname{Im}\rho_m \in \Big\{4 + \sqrt{2},\; 4 + \tfrac{5}{2\sqrt{2}},\; 4 + \tfrac{7}{2\sqrt{2}},\; 4 + 2\sqrt{2}\Big\},$$

each of which is at least $4 + \sqrt{2}$. □

Proof of Proposition 9.1. Fix $t \neq 0$, write $B = B_t(A')$, take Q_1, Q_2 from Lemma 9.3, and put $E = \|Q_2\|_2$ and $H = |Q_1|$. Then $H \geq 0$ is even, so $\widehat{H}(m)$ is real and $\widehat{H * H}(m) = \widehat{H}(m)^2 \geq 0$; moreover $\|H\|_1 \leq 3\int u = 3K$ and $\|H\|_2 \leq \|\widehat{\mathbf{1}_B}\|_2 + E = |B|^{1/2} + E$. We integrate (9.5) against $H * H \geq 0$. On the one hand, by Parseval and Lemma 9.4,

$$\begin{aligned} X &:= \int (\psi_t + 32K\mu)(H * H) = \sum_m \operatorname{Re}\rho_m \widehat{H}(m)^2 + 32K\mu\|H\|_1^2 \\ &\leq \Big(4 + \frac{1}{\sqrt{2}}\Big)\big(|B|^{1/2} + E\big)^2 + 288K^3\mu. \end{aligned}$$

On the other hand $H * H \geq |Q_1 * Q_1|$ pointwise, so by (9.5)

$$X \geq \int |\varphi_t|\,|Q_1 * Q_1| \geq \Big|\int \varphi_t (Q_1 * Q_1)\Big| = \Big|\sum_m i \operatorname{Im}\rho_m \widehat{Q_1 * Q_1}(-m)\Big|.$$

Since $Q_1 = \widehat{\mathbf{1}_B} - Q_2$ and $\widehat{\mathbf{1}_B} * \widehat{\mathbf{1}_B} = \widehat{\mathbf{1}_B}$, the coefficients of $Q_1 * Q_1$ are $\mathbf{1}_B(m) - 2\cdot\mathbf{1}_B(m)\widehat{Q_2}(m) + \widehat{Q_2}(m)^2$. The contribution of the first of these terms is $i\sum_{m\in -B}\operatorname{Im}\rho_m = i\sum_{m\in B}(-\operatorname{Im}\rho_m)$, of modulus at least $(4+\sqrt{2})|B|$ by Lemma 9.4; by the same lemma, Cauchy–Schwarz and Parseval, the other two contribute at most $9\big(2|B|^{1/2}E + E^2\big)$ in modulus. Comparing the two bounds for X and using $(4+\sqrt{2}) - (4+\frac{1}{\sqrt{2}}) = \frac{1}{\sqrt{2}}$,

$$\frac{|B|}{\sqrt{2}} \le \Big(13 + \frac{1}{\sqrt{2}}\Big)\big(2|B|^{1/2}E + E^2\big) + 288K^3\mu.$$

With $\alpha = 13\sqrt{2}+1 < 19.4$ and $\beta = 288\sqrt{2}\,K^3\mu < 408K^3\mu$ this reads $|B| \le 2\alpha E|B|^{1/2} + \alpha E^2 + \beta$, so $|B|^{1/2} \le \alpha E + (\alpha^2E^2 + \alpha E^2 + \beta)^{1/2}$ and $|B| \le (4\alpha^2 + 2\alpha)E^2 + 2\beta$. Since $E^2 \le 98K^3\mu$ by Lemma 9.3, we get $|B| \le \big(98(4\alpha^2+2\alpha) + 816\big)K^3\mu < 1.6\cdot 10^5 K^3\mu$. The second inequality of the proposition is $\mu \le 2K$. □

Remark 9.5. The proof uses only three pieces of information about F: the hypothesis $F \ge -K$, the identity $\int u = K$, and $\mu = \|F\|_1$ (through $\|T_1\|_1 = \|T_2\|_1 = \mu/2$ and $\|T_2\|_2^2 \le K\mu/2$). The three factors of K in the final bound enter as follows: one through $f_t \ge -4K$, which produces the constant $32K\mu$ in (9.5), and two through $\|H\|_1 \le 3\int u$; the terms in E^2 are of the same size. Consequently, within this argument, an improvement to $|B_t(A')| \ll K^{4-\delta}$ can only come from an upper bound $\mu \ll K^{1-\delta}$. Such a bound fails for the sets of Theorem 6.1, for which numerically $\mu \approx 0.97K$ (Section 10), and we do not know whether it holds for any set with $\lambda_A \le n^{1/2-\varepsilon}$.

9.2. **Sets with frequencies $O(n)$.**

Lemma 9.6 (Total boundary)**.** *Let $T_3(A') := \#\{(a,s) \in A'\times\mathbb{Z}_{>0} : a-s \in A',\ a+s\in A'\}$. Then $T_3(A') \le N^2/4$ and*

$$\sum_{t\ne 0}|B_t(A')| = N^2 - N - 2T_3(A') \ \ge\ \frac{N^2}{2} - N.$$

Proof. Write $r(t) = \#\{a\in A' : a-t\in A'\}$ and $r_3(t) = \#\{a\in A' : a-t\in A',\ a+t\in A'\}$, so that $|B_t(A')| = r(t) - r_3(t)$. Summing over $t\ne 0$, $\sum_{t\ne0} r(t) = \#\{(a,a')\in A'^2 : a\ne a'\} = N^2 - N$ and $\sum_{t\ne0} r_3(t) = \#\{(a,t) : t\ne 0,\ a\pm t\in A'\} = 2T_3(A')$, which gives the identity. For the bound on T_3, order $A' = \{a_1 < \cdots < a_N\}$. If (a_i, s) is counted by $T_3(A')$ then $a_i + s = a_j$ for some $j > i$ and $a_i - s = 2a_i - a_j = a_l$ for some $l < i$; since s determines j and l, the number of such s is at most $\min(N-i, i-1)$, and $\sum_{i=1}^{N}\min(N-i,i-1) \le N^2/4$. □

Proposition 9.7 (Frequencies $O(n)$)**.** *Let $C \geq 1$ and let $A \subset [1, Cn]$ be a set of $n \geq 2$ positive integers. Then*

$$\lambda_A^3 \|f_A\|_1 \;\geq\; \frac{n-1}{32CC_1} \qquad \text{and hence} \qquad \lambda_A \;\geq\; \Big(\frac{n-1}{64CC_1}\Big)^{1/4},$$

with C_1 the constant of Proposition 9.1.

Proof. Put $A' = A \cup (-A)$, so $N = 2n$, $K = 2\lambda_A$ and $\mu = 2\|f_A\|_1$. Proposition 9.1 gives $|B_t(A')| \leq 16C_1\lambda_A^3\|f_A\|_1$ for every $t \neq 0$. If $B_t(A') \neq \emptyset$ then $t \in (A'-A') \setminus \{0\} \subset [-2Cn, 2Cn] \setminus \{0\}$, so at most $4Cn$ values of t contribute to the sum in Lemma 9.6. Hence

$$2n^2 - 2n = \frac{N^2}{2} - N \;\leq\; \sum_{t \neq 0} |B_t(A')| \;\leq\; 4Cn \cdot 16C_1\lambda_A^3\|f_A\|_1,$$

which is the first claim; the second follows from $\|f_A\|_1 \leq 2\lambda_A$. □

Remark 9.8. (i) The proof shows more generally that every n-element set $A \subset \mathbb{N}$ satisfies $\max A \geq (n^2-n)/(32C_1\lambda_A^3\|f_A\|_1) \geq (n^2-n)/(64C_1\lambda_A^4)$. A set for which the exponent $1/5$ of [1, 14] is attained, that is $\lambda_A \leq n^{1/5+\varepsilon}$, would therefore need $\max A \geq cn^{6/5-4\varepsilon}$: the sets that could be extremal for Chowla's problem in the sense of having a lower bound worse than $n^{1/4}$ are sparse in their hull, whereas the sets constructed in this paper are dense in theirs. (ii) The selection of t in [1] (Lemma 7.4 there) and in [14] produces a shift with $|B_t(A')| \gg N/K$, which is what limits the general exponent to $1/5$; for sets with frequencies $O(n)$, Lemma 9.6 produces a shift with $|B_t(A')| \gg N$ instead, and the exponent $1/4$ results. (iii) For the sets $A(S)$ of Theorem 6.1 the bound is far from tight, since $-\min f_{A(S)} = \frac{1}{2}\sqrt{n} + O(n^{1/4}\sqrt{\log n})$; Proposition 9.7 says nothing about whether $\sqrt{n}$ is the true order for dense sets.

9.3. The L^1 norm.

Lemma 9.9 (Third moment of the positive part)**.** *Let $S_3(A')$ be the number of triples $(a,b,c) \in A'^3$ with $a+b+c=0$. Then*

$$S_3(A') \;\geq\; \frac{\big(N - K\mu/2\big)^2}{\mu/2} - \frac{K^2\mu}{2},$$

and if $N \geq 2K\mu$ then $S_3(A') \geq N^2/\mu$.

Proof. Expanding F^3 shows $S_3(A') = \int F^3 = \int F_+^3 - \int F_-^3$. By Cauchy–Schwarz,

$$\int F_+^2 = \int F_+^{1/2} F_+^{3/2} \leq \Big(\int F_+\Big)^{1/2} \Big(\int F_+^3\Big)^{1/2},$$

so $\int F_+^3 \ge (\int F_+^2)^2/\int F_+$, where $\int F_+ = \mu/2$ and $\int F_+^2 = N - \int F_-^2 \ge N - K\mu/2$ by (9.1); also $\int F_-^3 \le K^2\mu/2$ by (9.1). If $N \ge 2K\mu$ then $K\mu/2 \le N/4$, so the first term is at least $2(3N/4)^2/\mu = 9N^2/(8\mu)$, while $K^2\mu/2 = (K\mu)^2/(2\mu) \le N^2/(8\mu)$; the difference is at least N^2/μ. □

The lemma uses only $\int F = 0$, $\int F^2 = N$ and $F \ge -K$; the integrality of the coefficients enters through the interpretation of $\int F^3$ as a count. Since $\mu \le 2K$, it is never weaker than Roth's bound $S_3(A') \ge N^2/K - KN$ ([11, Lemma 5], [1, Lemma 4.1]) by more than a constant factor, and it is stronger by the factor K/μ. We combine it with the total-boundary identity of Shankar [14, Lemma 3.1], whose short proof we include.

Lemma 9.10 (Shankar)**.** *For every finite symmetric $A' \subset \mathbb{Z} \setminus \{0\}$,*

$$\sum_{t \in A'} |B_t(A')| \;\ge\; \frac{1}{3} S_3(A').$$

Proof. Let $P = A' \cap \mathbb{N}$, $Q = \#\{(u,v) \in P^2 : u+v \in P\}$ and $I = \#\{(u,v) \in P^2 : u - v \in A',\ u+v \in A'\}$. A solution of $a+b+c=0$ in A'^3 has no zero entry and not all entries of one sign, so it has exactly two positive entries or exactly two negative entries. In the first case the two positive entries form, in order of position, a pair $(u,v) \in P^2$ with $-(u+v) \in A'$, that is $u+v \in P$, and the position of the negative entry can be chosen in three ways; so there are $3Q$ such solutions, and by symmetry $3Q$ of the second kind: $S_3(A') = 6Q$. Next, with r and r_3 as in the proof of Lemma 9.6, $\sum_{t \in A'} r(t) = \#\{(a,t) \in A'^2 : a - t \in A'\} = S_3(A')$, via the bijection $(a,t) \mapsto (a, -t, t-a)$, and $\sum_{t \in A'} r_3(t) = \#\{(a,t) \in A'^2 : a - t,\ a + t \in A'\} = 4I$, because $(a,t) \mapsto (|a|,|t|)$ maps the pairs counted on the left onto the pairs counted by I (as $\{|a|-|t|, |a|+|t|\} \subset \{\pm(a-t), \pm(a+t)\}$ and $A' = -A'$) with fibres of size four. Hence $\sum_{t \in A'} |B_t(A')| = 6Q - 4I \ge 2Q$, since $I \le \#\{(u,v) \in P^2 : u+v \in A'\} = Q$. □

Theorem 9.11 (An L^1-sensitive lower bound)**.** *For every finite $A \subset \mathbb{N}$ with $|A| = n$,*

$$\lambda_A^3 \|f_A\|_1^2 \;\ge\; \frac{n}{48C_1},$$

with C_1 the constant of Proposition 9.1. Consequently $\lambda_A \ge (n/(192C_1))^{1/5}$, and if $\|f_A\|_1 \le \lambda_A^{1-\delta}$ for some $\delta \in (0,1]$ then $\lambda_A \ge (n/(48C_1))^{1/(5-2\delta)}$.

Proof. Let $A' = A \cup (-A)$, so $N = 2n$, $K = 2\lambda_A \ge 1$ and $\mu = 2\|f_A\|_1$. If $N < 2K\mu$, that is $n < 4\lambda_A \|f_A\|_1$, then $\lambda_A^3 \|f_A\|_1^2 = \lambda_A(\lambda_A\|f_A\|_1)^2 > \lambda_A n^2/16 \ge n/32 \ge n/(48C_1)$, using $\lambda_A \ge 1/2$. Otherwise Lemmas 9.9

and 9.10 and Proposition 9.1 give

$$\frac{2n^2}{3\|f_A\|_1} = \frac{N^2}{3\mu} \;\le\; \sum_{t\in A'} |B_t(A')| \;\le\; 2n\cdot C_1(2\lambda_A)^3\cdot 2\|f_A\|_1 = 32C_1 n\lambda_A^3\|f_A\|_1,$$

which is the claim. The consequences follow from $\|f_A\|_1 \le 2\lambda_A$ and from $\|f_A\|_1 \le \lambda_A^{1-\delta}$ respectively. □

Theorem 9.11 says that a set for which the exponent $1/5$ is close to sharp must have a large L^1 norm: if $\lambda_A \le n^{1/5+\varepsilon}$ then $\|f_A\|_1 \ge n^{1/5-3\varepsilon/2}/(48C_1)^{1/2}$, so its cosine sum is, in L^1, as large as its negative excursion up to a factor $n^{5\varepsilon/2}$. The known near-extremal sets for the upper bound behave in the same way: for the sets of Theorem 6.1, $\|f_A\|_1 = (0.48+o(1))\sqrt{n}$ numerically (Section 10), so $\|f_A\|_1 \approx \lambda_A$ and the theorem gives nothing beyond $1/5$ for them. Whether every set with $\lambda_A \le n^{1/2-\varepsilon}$ has $\|f_A\|_1 \asymp \lambda_A$ is a natural question that we cannot answer; a positive answer would not by itself improve the exponent, since Theorem 9.11 gains only from $\|f_A\|_1 \ll \lambda_A$.

The same substitution of $\int F_+$ for $\int(F+K)$ in a Cauchy–Schwarz step gives an L^1 form of Ruzsa's lemma [12, Lemma 3.1], in the formulation of [1, Lemma 4.2].

Lemma 9.12 (Ruzsa's lemma with the L^1 norm)**.** *Let $U, V \subset \mathbb{Z}\setminus\{0\}$ be finite with $|U| = |V|$, $U \subset A'$, $V \cap A' = \emptyset$ and $U - V \subset A'$. Then $|U| \le 4K\mu$; for $A' = A \cup (-A)$ this is $|U| \le 16\lambda_A\|f_A\|_1$.*

Proof. Let $g = \widehat{\mathbf{1}_U} - \widehat{\mathbf{1}_V}$. By Parseval, $\int gF = \#(U\cap A') - \#(V\cap A') = |U|$. Split $\int gF = \int gF_+ - \int gF_-$. Cauchy–Schwarz with the weight $F_+ \ge 0$ gives $|\int gF_+| \le (\int F_+)^{1/2}(\int |g|^2F_+)^{1/2}$, and $\int |g|^2F_+ = \int |g|^2F + \int |g|^2F_-$. Here $\int |g|^2F = \sum_{x,y\in U\cup V} \varepsilon_x\varepsilon_y \mathbf{1}_{A'}(x-y)$ with $\varepsilon = +1$ on U and -1 on V; the diagonal vanishes since $0 \notin A'$, the $U\times U$ and $V\times V$ terms contribute at most $|U|(|U|-1) + |V|(|V|-1)$, and each of the $2|U||V|$ cross terms equals -1 because $U - V \subset A' = -A'$. So $\int |g|^2F \le -2|U| \le 0$, and $\int |g|^2F_- \le K\int |g|^2 = 2K|U|$. Hence $|\int gF_+| \le (\frac{\mu}{2}\cdot 2K|U|)^{1/2} = (K\mu|U|)^{1/2}$. Also $|\int gF_-| \le \|g\|_2\|F_-\|_2 \le (2|U|)^{1/2}(K\mu/2)^{1/2} = (K\mu|U|)^{1/2}$. Therefore $|U| \le 2(K\mu|U|)^{1/2}$, that is, $|U| \le 4K\mu$. □

Corollary 9.13. *Every arithmetic progression contained in A has at most $64\lambda_A\|f_A\|_1 + 6$ terms.*

Proof. Let $\{y_0 + jd : 0 \le j < L\} \subset A$ with $d \ne 0$ and $L \ge 3$, put $p = \lfloor (L-1)/2\rfloor \ge 1$, $x_0 = y_0 + pd$ and $P = \{x_0 - pd, \dots, x_0 + pd\} \subset A \subset A'$; then $-P \subset A'$ as well. We follow Ruzsa [12] in the form given in [1, Lemma 4.4]. Let $U_0 = \{x_0 + d, \dots, x_0 + pd\}$ and $U_\ell = U_0 + \ell d$ for $\ell \ge 0$. Every element

of $U_0-(U_0-x_0)+\{0,d\}$ is of the form $x_0+(i-j+\epsilon)d$ with $1\le i,j\le p$ and $\epsilon\in\{0,1\}$, so $U_0-(U_0-x_0)+\{0,d\}\subset P$. Since $|U_0\cap A'|=p$ and $U_\ell\cap A'=\emptyset$ for large ℓ, there is $\ell\ge 0$ with $|U_\ell\cap A'|\ge p/2>|U_{\ell+1}\cap A'|$. If $|(U_\ell-x_0)\cap A'|<p/2$, put $U=U_\ell\cap A'$ and $V=(U_\ell-x_0)\setminus A'$; then $U-V\subset U_0-(U_0-x_0)\subset P$. Otherwise put $U=(U_\ell-x_0)\cap A'$ and $V=U_{\ell+1}\setminus A'$; then $U-V\subset(U_0-x_0)-U_0-d=-\big(U_0-(U_0-x_0)+d\big)\subset -P$. In both cases $U\subset A'$, $V\cap A'=\emptyset$, $U-V\subset A'$, $|U|\ge p/2$ and $|V|>p/2$. Removing 0 from V if it occurs there and discarding elements until $|U|=|V|$, we obtain sets as in Lemma 9.12 with $|U|=|V|\ge p/2-1$. Hence $p/2-1\le 16\lambda_A\|f_A\|_1$ and $L\le 2p+2\le 64\lambda_A\|f_A\|_1+6$. □

Ruzsa's bound $O(\lambda_A^2)$ is recovered from $\|f_A\|_1\le 2\lambda_A$. For an interval $A=[1,M]$ one has $\lambda_A\asymp M$ and $\|f_A\|_1\asymp\log M$, so Corollary 9.13 is sharp up to the logarithm for the interval, whereas Ruzsa's bound is off by a factor of M there.

10. Numerical verification

We implemented the construction of Sections 3 to 5 in Python (NumPy). The Singer set is obtained as the set of exponents $i\in\{0,\dots,v-1\}$ for which ϑ^i lies in a fixed two-dimensional subspace of $\mathbb{F}_{q^3}$ over $\mathbb{F}_q$, where ϑ is a primitive element of $\mathbb{F}_{q^3}$, and the perfect-difference property is verified directly. The tent thinning with parity block is sampled by rejection until $|S|=k$ exactly. At the sizes tested, the parameter r of Section 5 exceeds $q+1$ (it is designed for the asymptotic proof only), so the parity block was taken to be the $r=\lceil 3\sqrt{q}\rceil$ odd-offset elements closest to the centre; the seed of the random generator was 0 throughout. The minimum of f_A was located by a real FFT of the indicator of A on a grid of at least $16\max A$ points, followed by golden-section refinement of the 64 lowest grid values using the identity of Proposition 2.1. The identity itself was checked against the definition of f_A on a coarse grid, with agreement to 10^{-9} in all cases, and the absence of coincident frequencies was verified for every set. Table 1 lists the results.

Two further experiments illustrate the mechanism. At $q=401$, $k=80$, removing the parity block ($r=0$) gives $\min f_A=-55.93$, attained at $x=0.3376\pi$, that is, adjacent to the dangerous point $\pi/3$ of Zone II, with $\rho(x)^2<10^{-3}$ and $-g(x)/2=-15.9$; with $r=20,61,120$ the minimum rises to -54.25, -53.55, -52.67 and moves away from $\pi/3$. In every case the minimiser satisfies $\rho^2<1$, so the excess over $k/2$ is entirely the value of $g/2$ at a zero of ρ, as in Remark 8.5.

q	k	θ	r	n	$\max A$	$\min_x f_A$	$\dfrac{-\min f_A - k/2}{\sqrt{k}}$	$\dfrac{-\min f_A}{\sqrt{n}}$
61	12	0.749	24	132	7845	−9.639	1.051	0.839
101	20	0.665	31	380	26853	−15.689	1.272	0.805
199	40	0.635	43	1560	106144	−28.646	1.367	0.725
311	60	0.505	53	3540	265383	−42.141	1.567	0.708
401	80	0.522	61	6320	418031	−53.550	1.515	0.674
601	120	0.525	74	14280	1025874	−78.215	1.663	0.655
809	160	0.496	86	25440	1753844	−101.213	1.677	0.635

TABLE 1. Numerical minima of $f_{A(S)}$, $n = |A(S)| = k(k-1)$, for the construction with parity block $r = \lceil 3\sqrt{q} \rceil$ and random seed 0. The last column decreases towards $1/2$; the penultimate column grows slowly, consistent with an error term of order $\sqrt{k \log k}$.

The L^1 norm $\|f_A\|_1 = (2\pi)^{-1} \int_0^{2\pi} |f_A|$ of the same sets, computed as the mean of $|f_A|$ over the FFT grid, is 0.496, 0.490, 0.486, 0.484, 0.484, 0.483, 0.482 times $\sqrt{n}$ for the seven rows of Table 1. This is consistent with the identity of Proposition 2.1: away from the special points ρ behaves like a centred Gaussian with variance $k/2$, so $\rho^2 - k/2$ behaves like $\frac{k}{2}(Z^2-1)$ with Z standard normal, and $\mathbb{E}|Z^2-1| = 4\varphi(1) = 0.968$ gives $\|f_A\|_1 \approx 0.484k \approx 0.484\sqrt{n}$. Thus $\|f_A\|_1/\lambda_A \to 0.97$ along the construction, and Theorem 9.11 yields nothing beyond the exponent $1/5$ for these sets.

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